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Portfolio: <https://cornell-mae-ug.github.io/fa25-portfolio-mac624-stack/>

EDUCATION

Cornell University, Duffield College of Engineering

May 2028

Bachelor of Science in Mechanical & Aerospace Engineering

Ithaca, NY

- **GPA:** 4.0/4.33
- **Coursework:** Ansys FEA, Automotive Eng., Statics, Dynamics, Thermo, Linear Algebra, Differential Equations

The American School in London

June 2024

High School Diploma

London, England

- **GPA:** 3.97/4.0. **Honors:** National Honor Society, Honor Roll all Semesters, NMSC Commended Scholar
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EXPERIENCE

Brakes Subteam Member *Cornell Baja Racing SAE*

Nov. 2024 -

1st Place Iron Team. Arizona, Maryland, South Carolina 2025 Overall Win

- Design, manufacture, test entirely custom master cylinders, pedals & hydraulic brakes system.
- Utilize Ansys FEA to decrease weight of pedals by 27% and master cylinders by 30%.
- Change to separate piston-pushrod system and rigid mounting to eliminate piston binding.
- Spec new seal, bore machining, and anodization to reduce seal drag.
- Program & operate CNC mill and lathe to make all parts for master cylinders and pedals.
- Tune & numerically analyze brakes and further vehicle dynamic characteristics to maximize drivability.
- Introduce new brakes maintenance methods to ensure performance & consistency - zero competition failures in brakes on 2025 car.

Teaching Assistant – ENGRD 2020 Statics & Mechanics of Materials *Cornell Duffield Engineering*

**Fall 2025,
Spring 2026**

- Lead 21-person discussion section, teaching and reviewing new content.
- Assist 250 students in core curriculum work through office hours & online forum.
- Manage upkeep & usability of course webpage & assignment structure.

Design Lead, CAD/CAM Lead *Robotics - FIRST Team 1884 Griffins*

2021-2024

Founders' Award Recipient 2024, Engineering Inspiration Award 2023&2024

- Lead teams & held lectures in designing and prototyping a competitive robot.
- Qualified for World Championships 2023 & 2024.
- Introduced first ever CNC machining for the team, fitting bespoke cooling and enclosure.
- Educated younger team members on the importance of designing for manufacturing.
- Ensured quality & consistency while machining 75% of all parts for the competition robot.

Team Mentor, Event Coordinator *FIRST U.K. FTC London Tournament*

2022-2024

- Aided many new teams in preparation for their first tournaments.
- Administrated and scored multiple tournaments for 50+ teams in the greater London area.

Varsity Men's Rowing *The American School in London Boat Club*

2020-2024

- Raced bow seat in Henley Royal Regatta & Head of the Charles Regatta.

Custom Pen Plotter & Basic Software *Personal Project*

2023

- Designed and built successful pen plotting device and gCode generation software
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SKILLS

CAD/CAM (*SOLIDWORKS, Fusion 360, HSMWorks*), Programming (*MatLab, Java, Python, SQL, gCode*), Conversational CNC Lathe, Manual Mill & Lathe, ANSYS FEA, General Metal Fabrication, FDM 3D Printing, Skiing, Mending Textiles